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Research Article

Composition-dependent femtosecond laser damage and incubation effects in $Pb_2P_2O_7$–Nb_2O_5 phosphate glasses

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A B S T R A C T

Femtosecond laser micromachining was investigated in a series of phosphate glasses with systematically varied Nb_2O_5 concentration, aiming to correlate their linear optical properties with single-pulse and multi-pulse damage behavior. UV–Vis absorption spectroscopy revealed a progressive reduction in optical bandgap energy with increasing niobium oxide content, consistent with the formation of a more polarizable glass network. The bandgap values show excellent agreement with those previously reported for glasses of identical compositions, confirming the reliability of the structural–optical trends. Single-pulse and multi-pulse ablation thresholds were determined for all samples, demonstrating a monotonic decrease in damage fluence as Nb_2O_5 concentration increases. Incubation was observed in all glasses, with a systematic reduction of threshold fluence as the number of pulses increased, revealing a clear composition-dependent sensitivity to cumulative laser exposure. The incubation coefficients extracted from the fluence-versus-pulse-number analysis confirmed that samples with lower bandgap energies exhibit more pronounced multi-pulse modification behavior. To provide additional insight into the underlying ionization regime, the Keldysh parameter was calculated for each composition using measured bandgap energies and refractive index values obtained from the literature. All glasses exhibited $\gamma < 1$ under the experimental conditions, indicating a predominantly tunnelling ionization regime for the onset of damage. Together, these results establish a direct link between the optical properties of niobium phosphate glasses and their response to ultrafast laser irradiation, offering a consistent framework for predicting micromachining performance across this glass family.

1. Introduction

The development of advanced photonic devices relies on a wide range of material platforms, including organic [1,2] inorganic [3–6] and hybrid systems, each offering distinct advantages in terms of optical functionality, processing versatility, and integration potential. While organic materials are attractive due to their large optical nonlinearities, chemical tunability, and low-temperature processing, inorganic materials, including glasses, provide superior thermal stability, mechanical robustness, and long-term optical reliability [7]. The continuous expansion of photonic technologies has motivated research to identify materials that combine favorable optical performance with compatibility for precise micro- and nanofabrication.

Among these material classes, optical glasses occupy a critical position in photonics due to their broad transparency windows, compositional flexibility, and ability to incorporate a wide variety of functional dopants. Glasses enable fine control of linear and nonlinear optical properties through compositional design, while maintaining excellent homogeneity and scalability [8,9]. The realization of practical photonic devices depends on reliable microfabrication techniques capable of producing well-defined structures with high spatial resolution and minimal collateral damage. In this context, laser-based microfabrication, especially using femtosecond pulses, has emerged as a powerful tool for structuring glasses and other materials, enabling the fabrication of waveguides, microresonators, and complex three-dimensional photonic architectures [10–12].

Femtosecond (fs) laser micromachining is as a versatile and powerful technique for high-precision material processing, enabling the creation of micro- and nanostructures with exceptional spatial resolution and minimal collateral damage [13,14]. Due to the ultrashort pulse duration and high peak intensities, fs-laser irradiation triggers nonlinear absorption processes that confine energy deposition to the focal volume, effectively reducing thermal diffusion and undesired melting [15–17]. This capability makes fs-laser processing highly attractive for applications ranging from integrated photonic circuits to microfluidic platforms, where precise control of material modification is essential [18–21].

An essential aspect of fs-laser micromachining is its sensitivity to

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both irradiation parameters and the material's intrinsic properties. Pulse duration, wavelength, fluence, and the number of incident pulses define the extent of material modification, while multiple-pulse irradiation can induce significant incubation effects, lowering the damage threshold as structural changes accumulate within the focal volume [22–24]. Understanding how the composition and physical properties of materials influence these mechanisms is crucial for optimizing microfabrication strategies and tailoring material response for specific applications. In addition to subsurface modification and buried waveguide fabrication, surface femtosecond laser ablation also plays an important role in photonic device engineering. Controlled surface micromachining enables the fabrication of microstructured interfaces, diffraction gratings, optical coupling elements, and microfluidic channels, as well as surface texturing for light extraction and sensing applications [25–27]. Moreover, accurate knowledge of surface damage thresholds and incubation behavior is essential for defining safe and reliable processing windows, even when targeting subsurface photonic structures [28].

Phosphate-based glasses have long attracted attention in photonics due to their good optical transparency in the UV–visible–near-infrared region, rare-earth solubility, and ability to incorporate heavy metal oxides [29,30]. When combined with metal oxides such as Nb_2O_5, WO_3, or Bi_2O_3, phosphate based glasses exhibit enhanced chemical durability, increased refractive indices, and strong nonlinear optical responses [8]. Among these systems, lead pyrophosphate–niobium oxide ($Pb_2P_2O_7$–Nb_2O_5) glasses stand out as promising candidates for photonic applications [31,32].

The incorporation of Nb_2O_5 into the $Pb_2P_2O_7$ glass network promotes the formation of highly coordinated NbO_6 units, which connect to phosphate tetrahedra through strong P–O–Nb bonds [33–35]. This structural reinforcement increases the glass transition temperature (T_g), improves mechanical stability, and significantly enhances the refractive index [8]. Moreover, the presence of Nb^{5+}, with their empty d orbitals, contributes to a high third-order nonlinear optical susceptibility, making these glasses suitable for applications in nonlinear optics, optical switching, and high-performance waveguiding components [36]. Previous studies have demonstrated their photonic potential, showing high transparency, strong nonlinear responses, and suitability for the fabrication of optical fibers and other integrated components [8,36].

However, despite the attractive thermal and optical properties of $Pb_2P_2O_7$–Nb_2O_5 glasses, their response to fs-laser irradiation remains poorly understood. Thus, we examine how intrinsic material parameters, such as bandgap energy and refractive index, which systematically vary with Nb_2O_5 content, influence the ablation threshold under ultrafast irradiation. The correlations discussed here are based on compositional trends and macroscopic laser–matter interaction outcomes, without direct structural characterization of the irradiated regions. Establishing these relationships is an important step toward assessing the feasibility of integrating these materials into fs-laser-based microfabrication processes and for guiding future investigations focused on the structural mechanisms involved.

In this work, we systematically investigate fs-laser micromachining in a compositional series of $Pb_2P_2O_7$–Nb_2O_5 phosphate glasses, aiming to establish direct correlations between glass composition, optical properties, and laser-induced damage behavior. Six distinct glass compositions were studied, spanning a broad range of Nb_2O_5 concentrations, enabling consistent assessment of single-pulse and multi-pulse damage thresholds and incubation effects. By combining optical bandgap analysis with femtosecond laser ablation experiments, this study provides a comprehensive framework for understanding how compositional tuning changes the micromachining response of niobium-containing phosphate glasses under ultrafast laser irradiation.

2. Experimental

2.1. Bulk glass synthesis

The glass samples of the $(100-x)Pb_2P_2O_7 - xNb_2O_5$ system, with x = 10, 20, 30, 40, 50 60 mol%, were prepared using the conventional melt-quenching technique, following the procedure described in Ref. [8]. High-purity Nb_2O_5 (99.8%, Aldrich) and lead orthophosphate ($PbHPO_4$) were used as starting reagents. $PbHPO_4$ was synthesized via precipitation of a lead salt solution with orthophosphoric acid at room temperature, as detailed in Refs. [8,37].

The weighed batches were first heated at 200°C for 1 h in air to remove residual moisture and gases, since $PbHPO_4$ is obtained via a wet-chemical process. The mixtures were then melted in open air at temperatures ranging from 950 °C to 1200 °C, depending on the Nb_2O_5 concentration, and kept at these temperatures for 40 min to ensure homogenization and complete fining. Subsequently, the melts were poured into a stainless steel mold preheated at 20°C below T_g of each composition, followed by an annealing step at the same temperature for 2 h to minimize internal stresses caused by thermal cooling. The resulting bulk glasses exhibited high optical quality and were polished to obtain flat, parallel and smooth surfaces suitable for femtosecond laser micromachining experiments.

2.2. Femtosecond laser micromachining

The microfabrication of the PxNy samples was performed using a diode-pumped Yb:KGW femtosecond laser system, delivering ultrashort pulses with a duration of approximately 216 fs at 1030 nm. The system provides an adjustable repetition rate ranging from 20 Hz to 200 kHz, enabling the exploration of different laser–matter interaction regimes.

The laser beam was focused onto the sample surface using a microscope objective with a numerical aperture (NA) of 0.65, ensuring high spatial resolution and precise energy deposition, which are essential for obtaining well-defined microstructures. The samples were mounted on a computer-controlled, motorized three-axis (xyz) translation stage with submicron resolution, enabling highly accurate and reproducible positioning during the microfabrication process. The fabrication process was monitored in real time using a CCD camera aligned with the optical axis of the focusing objective, providing visual feedback on the laser–sample interaction and enabling precise control of the microstructure formation. The laser beam exhibited a near-Gaussian spatial profile and linear polarization, which was kept constant throughout all experiments. A schematic representation of the experimental setup described is shown in Fig. 1.

During the experiments, the pulse energy was varied from 65 nJ to 615 nJ, while the number of pulses per spot was controlled by adjusting both the repetition rate (20–200 kHz) and the scanning speed (12.5–25 μm/s). The number of pulses per spot, or pulse overlap (N) was calculated according to $N \approx 1.25 \times f \times {w_0}/{v}$, where f is the repetition rate, w_0 is the beam waist radius, and v is the scanning speed. Thus, N was carefully controlled, spanning from N = 1 up to approximately N ≈ 10000. For each fixed N, the pulse energy was systematically varied and the damage threshold fluence was independently determined from the linear fit of the squared line radius (r^2) as a function of pulse energy. Therefore, each incubation point corresponds to an independent r^2 versus energy analysis. This approach allowed us to obtain the incubation curve, which describes the dependence of the ablation threshold on the accumulated number of pulses.

Since the samples exhibit different compositions of $Pb_2P_2O_7$ and Nb_2O_5, the influence of composition on the laser processing response was systematically investigated. This comparative analysis is particularly relevant because both the T_g and the nonlinear optical effects reported for similar compositions are expected to play a significant role in determining the material's behavior under ultrafast laser irradiation.

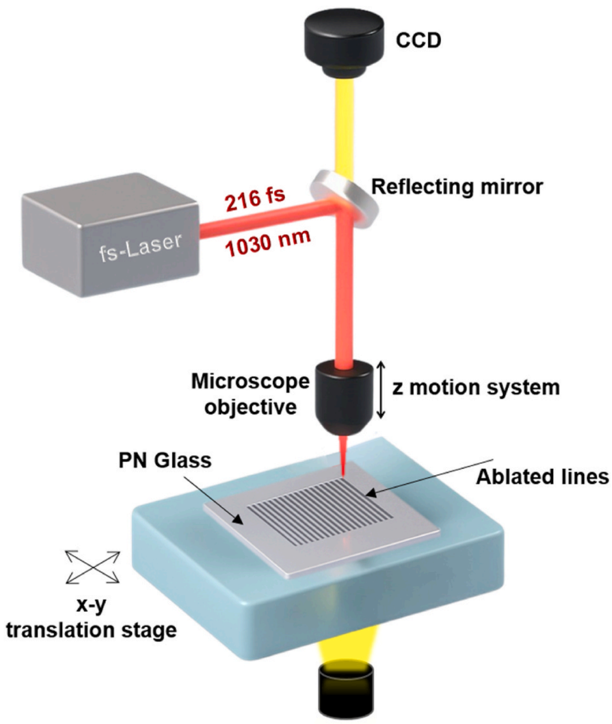

Fig. 1. Experimental setup for femtosecond laser micromachining.

The fabricated microstructures were systematically characterized to assess both their morphological and optical properties. Scanning Electron Microscopy (SEM) was performed using a FEI Inspect-F50 microscope to obtain high-resolution images of the laser-processed regions, enabling detailed analysis of surface features and structural modifications induced by femtosecond laser micromachining. Additionally, UV-Vis absorption spectroscopy was performed on a Shimadzu UV-1800 spectrophotometer to evaluate the optical response of the glasses. The acquired spectra were used to estimate optical band gap energies and to infer variations in the linear refractive indices across different glass compositions, providing complementary insights into the relationships among glass structure, composition, and laser-matter interaction.

3. Results and discussion

Fig. 2 shows the UV–Vis absorption spectra of the six investigated

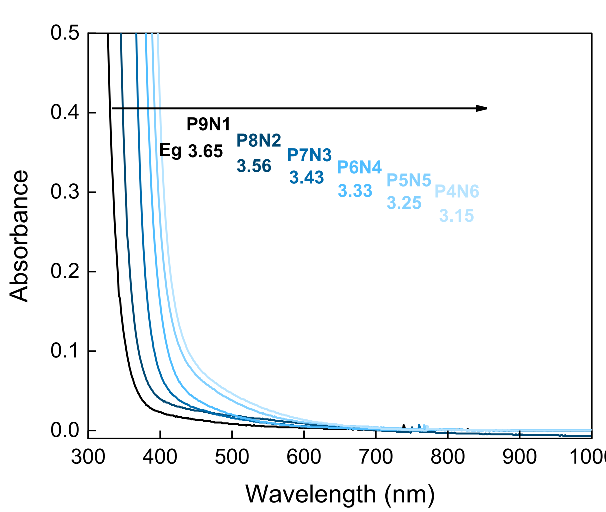

Fig. 2. – UV–Visible absorption spectra of the $Pb_2P_2O_7$–Nb_2O_5 glass samples.

$Pb_2P_2O_7$–Nb_2O_5 glass samples. A systematic redshift of the absorption edge is observed as the Nb_2O_5 content increases. Samples with lower concentration (P9N1 and P8N2) exhibit absorption edges at shorter wavelengths, whereas those with higher Nb_2O_5 fractions (P5N5 and P4N6) display progressively longer-wavelength absorption edges, indicating a clear modification of the electronic structure with composition.

The optical bandgap (E_g) was estimated using the Tauc method, assuming an indirect allowed transition. The obtained values ranged from 3.65 eV for the P9N1 sample to 3.15 eV for P4N6, with a monotonic decrease with increasing Nb_2O_5 content. This behavior can be attributed to structural modifications in the glass network induced by Nb_2O_5 incorporation. At lower concentrations, the $Pb_2P_2O_7$ matrix dominates, which is associated with higher bandgap energies. As Nb_2O_5 content increases, the lead-phosphate chains are modified and, to some extent, niobium acts as a network former, introducing additional nonbridging oxygens and a greater semiconductor character [38], thus lowering the optical bandgap.

This trend is consistent with previous studies on niobium-containing phosphate glasses, where the addition of Nb_2O_5 not only decreases E_g, but also significantly enhances the linear and nonlinear refractive index [8,39]. A clear inverse correlation between E_g and n is observed. Samples with higher Nb_2O_5 content exhibit lower bandgap energies and, consequently, larger refractive indices.

Understanding these optical properties is crucial for interpreting the interaction of these materials with femtosecond laser pulses. The E_g and n directly influence key aspects of light–matter coupling, including multiphoton absorption, the critical power for self-focusing, and the spatial confinement of energy deposition during laser microfabrication [40,41]. In particular, higher refractive index samples are expected to exhibit stronger nonlinear responses, which could lead to lower energy thresholds for permanent structural modification under ultrashort pulse irradiation [42]. Therefore, the trends revealed in Fig. 2 provides an essential foundation for the subsequent analysis of femtosecond laser micromachining, offering insight into how compositional tuning of the $Pb_2P_2O_7$–Nb_2O_5 glass system governs its optical response and processing window.

Femtosecond laser micromachining experiments were performed on all six glass samples (PxNy), and Fig. 3 summarizes the representative results obtained for the P4N6 composition. Similar behaviors were observed for the other samples. In Fig. 3a, SEM images show microstructures fabricated with $N = 10000$ pulses per spot, with pulse energy varying from 65 nJ up to 165 nJ. As the pulse energy increases, the diameter of the ablated regions grows significantly, indicating a strong correlation between deposited energy and the extent of material removal, as demonstrated in other studies [43–45]. Similarly, in Fig. 3b, SEM images display the results of single-pulse ($N = 1$) microfabrication, where the pulse energy was varied from 125 nJ to 615 nJ. Compared to the multipulse regime, higher energies are required to produce visible features under single-shot irradiation, and the morphology of the ablated spots is notably different, exhibiting cleaner edges and less collateral damage.

To quantify these observations, the squared radius r^2 of the fabricated microstructures was plotted as a function of pulse energy. Fig. 3c presents the experimental data for $N = 10000$, corresponding to the SEM images of Fig. 3a. By fitting the data using the zero-damage method [46], we determined a beam waist radius of $w_0 = 1.85$ μm and a threshold energy of $E_{th} = 48$ nJ. In this regime, the radius varied from approximately 0.81 to 1.45 μm (±0.07 μm).

Similarly, Fig. 3d shows the same analysis for the single-pulse case (Fig. 3b). Here, the extracted values were $w_0 = 1.24$ μm and $E_{th} = 64$ nJ, with the feature radius varying between 0.69 μm and 1.30 μm. These results highlight the higher energy required to initiate material removal under single-pulse irradiation compared to the multipulse regime.

The calculated threshold fluences were $F_{th} = 0.36$ J/cm^2 for 10000 pulses and $F_{th} = 1.48$ J/cm^2 for single-pulse irradiation, evidencing a clear incubation effect when increasing the number of pulses per spot.

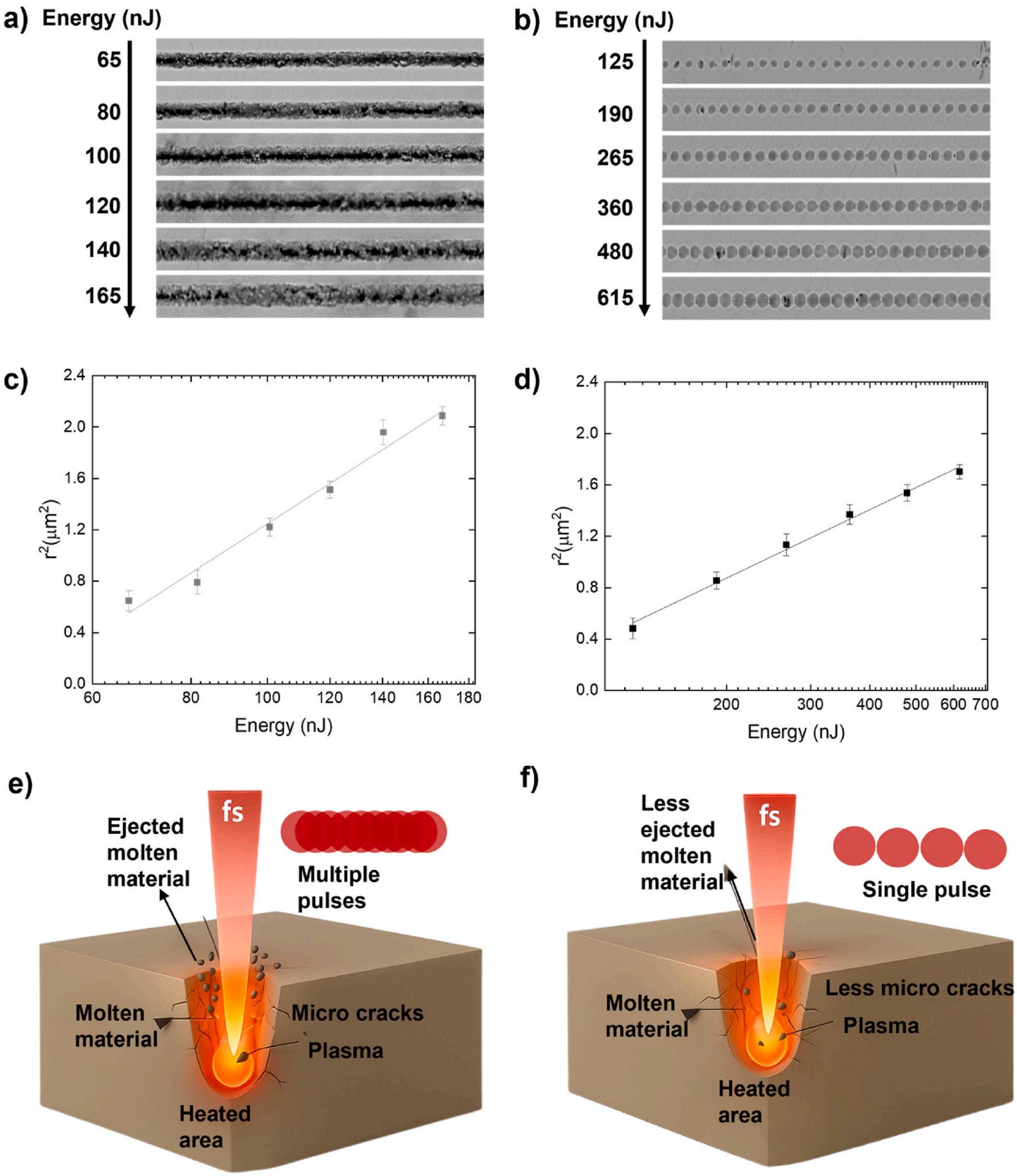

Fig. 3. – Femtosecond laser micromachining of the P4N6 glass. a) SEM images of microstructures fabricated using N = 10,000 pulses per spot. b) SEM images of microstructures fabricated using N = 1 pulse per spot. (c,d) Squared radius (r^2) as a function of pulse energy fitted using the zero-damage method. (e,f) Schematics illustrating the multipulse and single-pulse interaction regimes.

To further illustrate the distinct micromachining regimes, Fig. 3e and f schematically depict the interaction mechanisms for multipulse and single-pulse fabrication, respectively. In the multipulse regime, cumulative heating leads to larger molten volumes, enhanced plasma generation, and the formation of microcracks, resulting in greater material ejection and less-defined edges. Conversely, the single-pulse regime involves localized energy deposition, reduced thermal accumulation, cleaner ablation profiles, and fewer microcracks [47,48], consistent with the SEM observations.

Fig. 4 summarizes the role of incubation effects during femtosecond laser micromachining of the Nb_2O_5-containing glass samples. The presented data correspond to sample P4N6 as a representative example; however, identical experiments were systematically conducted for all six compositions, and a consistent behavior was observed.

In Fig. 4a, SEM images show the evolution of the microstructures fabricated with a constant single-pulse energy of 125 nJ, while varying the number of incident pulses per spot (N = 1, 5, 10, 50, 100, 1000, 5000, and 10000). The morphological transformations are strongly dependent on N. For a single pulse, only superficial modifications and surface melting are visible, consistent with the energy being close to the

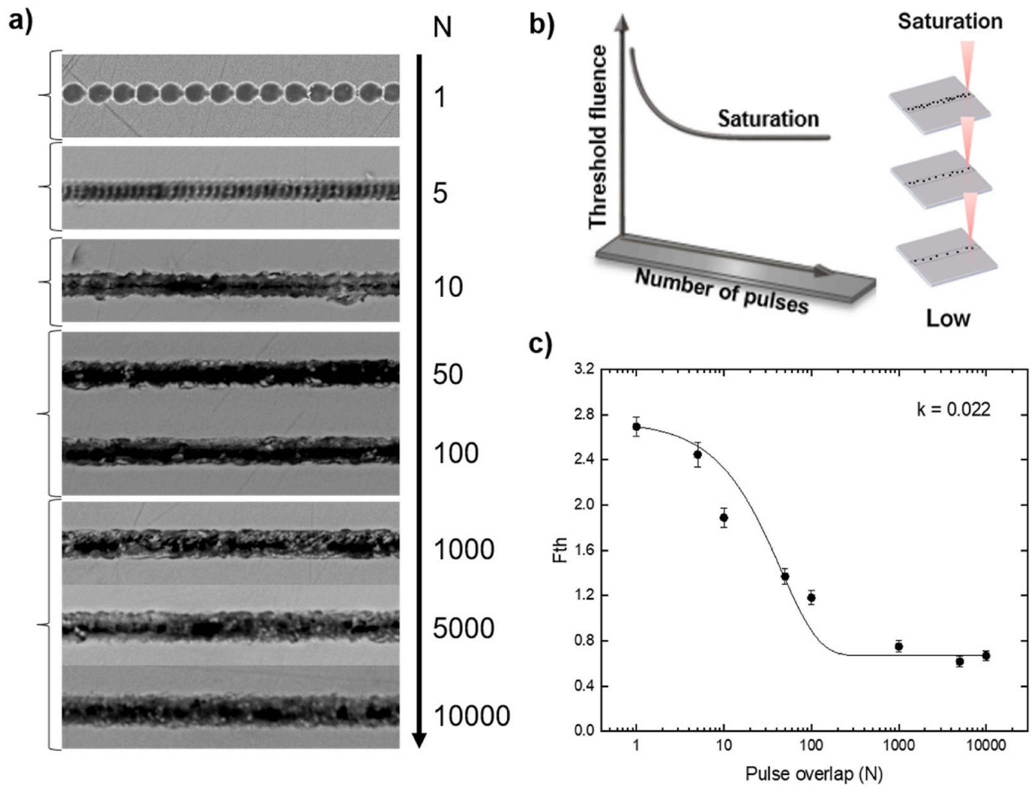

Fig. 4. Incubation effects during femtosecond laser micromachining of the P4N6 glass. a) SEM images of microstructures fabricated with a constant pulse energy of 125 nJ and a varying number of pulses. b) Schematic illustration of the defect-accumulation mechanism responsible for the reduction of the ablation threshold under multipulse irradiation. c) Experimentally measured ablation threshold fluence as a function of pulse number, fitted using the exponential incubation model.

ablation threshold. As N increases, a progressive accumulation of defects promotes more efficient light absorption, resulting in larger diameters. For $N \geq 1000$, the microstructures exhibit significant material ejection and the onset of redeposited debris around the rims, indicating a transition to a regime dominated by nonlinear absorption and cumulative energy deposition.

To provide a conceptual framework for this behavior, Fig. 4b presents a schematic illustration of the incubation process. Repeated pulse irradiation progressively increases the defect density within the material, locally enhancing optical absorption and lowering the effective threshold fluence [49,50]. [51]. Subsequent pulses encounter a progressively weaker lattice due to accumulated dangling bonds, free carriers, and local bond breaking, facilitating further structural modifications. Once a critical defect density is reached, the system enters a saturation regime, where additional pulses do not produce a significant reduction in the ablation threshold [52]. This trend is quantitatively captured in Fig. 4c, which plots the experimentally determined ablation threshold fluence (F_{th}) as a function of the number of pulses for P4N6. The data were fitted using the exponential defect accumulation model

$$F_{th,N} = (F_{th,1} - F_{th,\infty})e^{-k(N-1)} + F_{th,\infty} \quad (1)$$

where $F_{th,1}$ is the single-pulse ablation threshold fluence, $F_{th,\infty}$ is the saturation fluence for an infinite number of pulses, and k is the incubation parameter that characterizes the rate of defect accumulation.

For P4N6, the fitting yielded $F_{th,1} \approx 1.48$ J/cm^2 and $F_{th,\infty} \approx 0.36$ J/cm^2, showing a significant decrease in the threshold fluence as the number of pulses increases from 1 to 10000. This pronounced reduction highlights the strong cumulative effects in the material. The incubation parameter for P4N6 was determined to be $k \approx 0.02$, indicating a moderate defect accumulation rate.

When comparing the six studied samples, we observed that the incubation parameter increases slightly with higher Nb_2O_5 concentration. Samples with lower Nb_2O_5 content, such as P9N1 and P8N2, exhibited lower k values, meaning that a larger number of pulses is required to reach significant defect accumulation. Conversely, samples richer in Nb_2O_5, such as P5N5 and P4N6, show a slightly higher k, reflecting a greater sensitivity to multipulse irradiation. This correlation can be explained by the impact of Nb_2O_5 on the linear and nonlinear refractive indices of the glass matrix. As demonstrated in the optical characterization (Fig. 2), higher Nb_2O_5 content leads to higher refractive indices, which enhance local field confinement and multiphoton absorption, thereby accelerating defect accumulation under repeated femtosecond pulses [53]. Interestingly, despite this trend, the variation in k across the six samples remains moderate, suggesting that the dominant mechanism of incubation is not purely compositional but also strongly dependent on the intrinsic defect formation dynamics of the glass network. In all cases, the observed behavior is consistent with previously reported studies on dielectrics and semiconductors irradiated with ultrashort pulses [54–56].

From an applications standpoint, understanding the incubation behavior is essential for optimizing femtosecond laser microfabrication. By tailoring the pulse number and energy, one can exploit the incubation regime to achieve controlled ablation with sub-micrometer resolution while minimizing collateral damage and redeposition effects. Conversely, when precise surface modification without material removal is desired, operating below the single-pulse threshold and avoiding defect accumulation is critical.

To gain deeper insight into the ionization mechanisms underlying femtosecond laser interaction with $Pb_2P_2O_7$–Nb_2O_5 glasses, the Keldysh parameter (γ) was calculated for the single-pulse damage thresholds across the compositional series using

$$\gamma_K = \frac{\omega}{e}\sqrt{\frac{(m\varepsilon_0 c n_0 E_g)}{I_0}} \tag{2}$$

where ω is the laser angular frequency, e and m are the electron charge and reduced mass, ε_0 is the vacuum permittivity, c is the speed of light, n_0 is the linear refractive index, E_g is the optical bandgap, and I_0 is the laser intensity at threshold [57]. These γ values are presented in Fig. 5a as a function of Nb_2O_5 molar concentration, increasing modestly from 0.4 (P9N1) to 0.45 (P4N6). All values lie well below unity, consistent with the tunneling ionization regime ($\gamma < 1$), confirming that even for the composition with the highest damage sensitivity, the process remains dominated by strong-field tunneling rather than multiphoton absorption [58]. [59]. In fused silica, experimental and simulation studies demonstrate that multiphoton ionization become significant only when γ approaches or exceeds unity, whereas for comparable γ values (<0.5), tunneling remains the primary ionization pathway [59]. The consistency across different oxides supports the universality of this regime under high-field, ultrashort-pulse conditions.

Fig. 5b offers a conceptual representation of the ionization regimes: at $\gamma < 1$, tunneling dominates; $\gamma \approx 1$ marks the intermediate regime; and $\gamma > 1$ signifies the onset of multiphoton ionization. The experimental γ range plotted for all six glass compositions falls within the tunneling domain. This is significant because tunneling provides highly localized energy deposition [28,60], which is known to produce clean, smooth ablation profiles with minimal collateral damage, consistent with the microstructural morphology observed in Fig. 3.

Furthermore, because tunneling efficiency depends strongly on material parameters (E_g, n_0) and laser field strength [61], the subtle compositional trend in γ mirrors the measured variation in single-pulse damage thresholds and ablation morphology. The P4N6 composition, with the lowest threshold and largest γ among the series, reflects a slight but consistent shift toward stronger field coupling due to its inherent optical properties. The γ analysis, together with the schematic in Fig. 5b, reinforces that single-pulse femtosecond damage in these $Pb_2P_2O_7$–Nb_2O_5 glasses proceeds via tunneling ionization, and that compositional tuning moderately shifts the sensitivity but does not fundamentally alter the ionization class.

Fig. 6 summarizes the evolution of the ablation threshold fluence, F_{th}, as a function of the number of pulses for the six studied samples. Experimental points are plotted for N = 1, 5, 10, 100, and 1000, allowing a direct comparison between different Nb_2O_5 concentrations. For single-pulse irradiation (N = 1) it is possible to observe that F_{th} decreases as the Nb_2O_5 content increases. The highest threshold fluence is measured for sample P9N1, $F_{th,1} \approx 2.08$ J/cm^2, which contains the lowest Nb_2O_5 concentration, whereas sample P4N6, with the highest

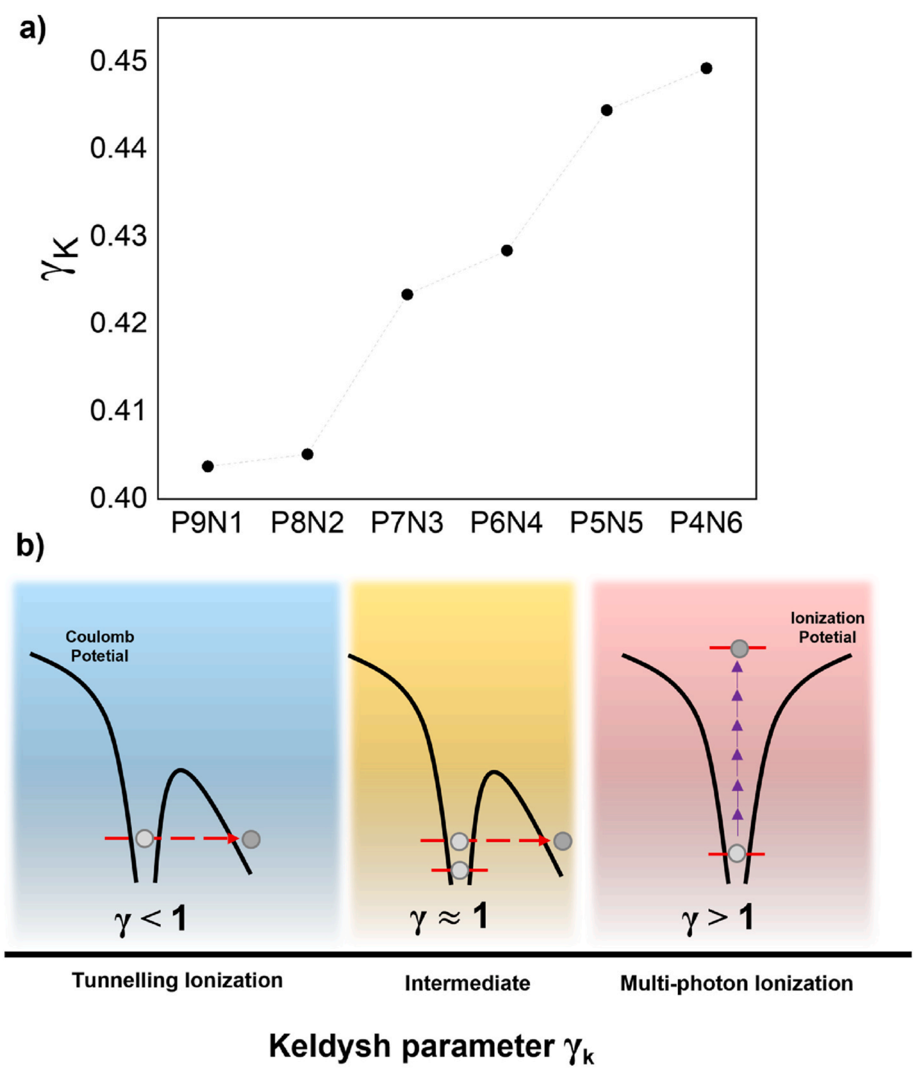

Fig. 5. Keldysh parameter and ionization regime for the $Pb_2P_2O_7$–Nb_2O_5 glasses.
a) Calculated Keldysh parameter (γ) for the six glass compositions based on the single-pulse damage thresholds. b) Schematic illustration of the ionization regimes defined by γ.

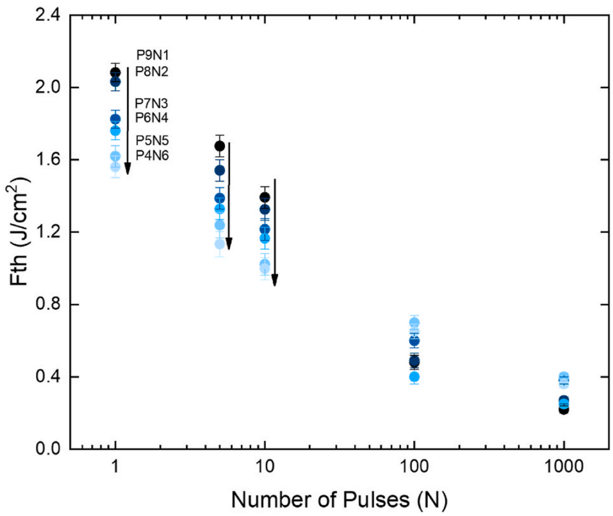

Fig. 6. – The evolution of the ablation threshold fluence as a function of the number of pulses for the $Pb_2P_2O_7$–Nb_2O_5 glass samples.

Nb_2O_5 content, shows the lowest single-pulse threshold, F_{th},1 ≈ 1.56 J/cm^2. When increasing the number of pulses to N = 5 and N = 10, the same compositional dependence persists; samples richer in Nb_2O_5 still exhibit lower F_{th} compared to those with less Nb_2O_5. For instance, at N = 10, F_{th} decreases from approximately 1.39 J/cm^2 in P9N1 to 1.00 J/cm^2 in P4N6, representing a ∼28% reduction.

However, for higher numbers of pulses (N = 100 and N = 1000), the trend becomes less pronounced. For example, at N = 1000, F_{th} values converge within a narrow range, from 0.22 J/cm^2 (P9N1) to 0.40 J/cm^2 (P5N5). This convergence indicates that at sufficiently high pulse counts, the ablation process is dominated by defect accumulation rather than intrinsic material properties. Once a critical density of defects is reached, differences in compositions become negligible, and the system approaches a saturation regime for all samples. This behavior reinforces the key role of incubation effects in fs-laser processing of these glasses. At low N, material-specific properties such as bandgap energy and refractive index govern the ablation threshold. In contrast, at high N, the process is dictated by nonlinear defect generation, multiphoton ionization, and avalanche ionization, leading to nearly composition-independent thresholds.

Interestingly, despite the convergence at high N, slight residual differences persist. Samples with higher Nb_2O_5 concentrations (P4N6 and P5N5) tend to retain slightly higher Fth values even after 1000 pulses. This could be attributed to local structural effects induced by Nb incorporation, such as changes in the glass network rigidity or defect formation energies, which subtly influence the defect accumulation dynamics.

From an applications perspective, these findings are particularly relevant for high-precision microfabrication. For processes requiring minimal thermal damage and precise control of ablation depth, operating in the low regime of N is advantageous, as compositional differences can be exploited to tailor processing windows. Conversely, for bulk modification or deep structuring, using high N allows the process to leverage defect accumulation for efficient energy coupling across all compositions.

For instance, studies on borosilicate glass under femtosecond irradiation reported a decrease in F_{th} from 1.94 J/cm^2 for single pulses to 0.87 J/cm^2 for N = 100, representing a reduction of about 55%, attributed to defect accumulation and structural relaxation [62]. Similarly, experiments on fused silica demonstrated that the threshold fluence decreases sharply with increasing pulse number, with a reported single-pulse threshold of 6.23 ± 0.23 J/cm^2 and an incubation parameter of k ≈ 0.058, before reaching a stable value at higher pulse counts [63]. In tellurite-based glasses a pronounced incubation effect is observed, single-pulse threshold fluences on the order of ∼ 6–8 J/cm^2 progressively decrease to values near 1.0 J/cm^2 as the pulse number increases, consistent with incubation parameters of approximately k ≈ 0.04–0.05 [64].

The similarity between our results and these previous studies highlights the universality of incubation phenomena under femtosecond laser irradiation. Although the absolute values of F_{th} vary depending on the material's bandgap, structure, and composition, the overall trend is consistent, increasing the number of pulses lowers the damage threshold until a saturation regime is reached. Notably, the $Pb_2P_2O_7$–Nb_2O_5 glass system studied here exhibits threshold reductions comparable to those reported for other oxide glasses and crystalline materials, demonstrating that defect-assisted absorption plays a dominant role in the laser–material interaction. Moreover, slight compositional differences among our samples modulate the magnitude of F_{th}, suggesting that tailoring Nb_2O_5 concentration provides a route to tune the resistance of these glasses to ultrafast laser processing. These findings not only corroborate existing theoretical and experimental frameworks for incubation effects but also provide new insights into the role of glass composition in defining femtosecond micromachining thresholds. By quantifying both single- and multipulse thresholds across a compositional series, this work contributes valuable data for optimizing laser parameters in applications such as precise microstructuring, waveguide fabrication, and photonic device integration.

4. Conclusions

In this work, fs-laser micromachining was systematically investigated in a series of $Pb_2P_2O_7$–Nb_2O_5 phosphate glasses with varying Nb_2O_5 content, aiming to establish correlations between composition, optical properties, and laser-induced damage behavior. UV–Vis spectroscopy revealed a monotonic reduction of the optical bandgap with increasing Nb_2O_5 concentration, consistent with enhanced network polarizability and increased refractive index. These compositional trends were reflected in the fs-laser response, with both single-pulse and multi-pulse ablation thresholds decreasing as the Nb_2O_5 content increased. Incubation effects were observed for all glass compositions, leading to a significant reduction of the damage threshold with increasing pulse number. The extracted incubation parameters indicate that glasses with lower bandgap energies exhibit enhanced sensitivity to cumulative laser exposure, while at high pulse numbers the ablation thresholds converge toward a saturation regime dominated by defect accumulation rather than intrinsic material properties. This behavior highlights the transition from composition-controlled to defect-controlled laser–matter interaction as the number of pulses increases. Analysis of the Keldysh parameter for single-pulse damage thresholds showed that all investigated compositions operate in the tunneling ionization regime under the present experimental conditions, with only modest variations across the compositional series. This confirms that the observed differences in laser damage thresholds primarily arise from compositional modulation of the electronic structure and local field enhancement, rather than a fundamental change in the ionization mechanism.

Overall, the results demonstrate that Nb_2O_5 concentration provides an effective parameter for tuning the femtosecond laser micromachining response of $Pb_2P_2O_7$-based phosphate glasses. The established correlations between optical bandgap, incubation behavior, and damage thresholds offer valuable guidelines for optimizing ultrafast laser processing conditions and support the potential of these glasses for photonic microfabrication applications, including precision structuring and integrated optical device fabrication.

CRediT authorship contribution statement

Kelly T. Paula: Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Vinicios T.R. Tranzil:** Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Leandro O.E. Silva:** Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. **Danilo Manzani:** Conceptualization, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **Cleber R. Mendonça:** Conceptualization, Formal analysis, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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